Practice workbook · Trace the route of blood through the heart, lungs and body

Ordering chambers, vessels and organs blood passes through, including the double-pump idea and arrows on diagrams.

9 question parts: 1 worked-example retry and 8 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: HBI-TRACE-DOUBLE-CIRCULATION-ROUTE

1. Question 5 · part a · 5 marks

June 2023 Paper 2

Worked-example retry · Attempt part a only.

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5 (a) A red blood cell travelling from the liver to the kidney passes through six different arteries and veins.

Complete the flow chart by naming these blood vessels in order. The first one has been done for you.

(5)

LIVER

↓

hepatic vein

↓

.....

↓

.....

↓

.....

↓

.....

↓

.....

↓

KIDNEY

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P 7 2 5 8 7 A 0 1 3 2 4

Turn over ▶

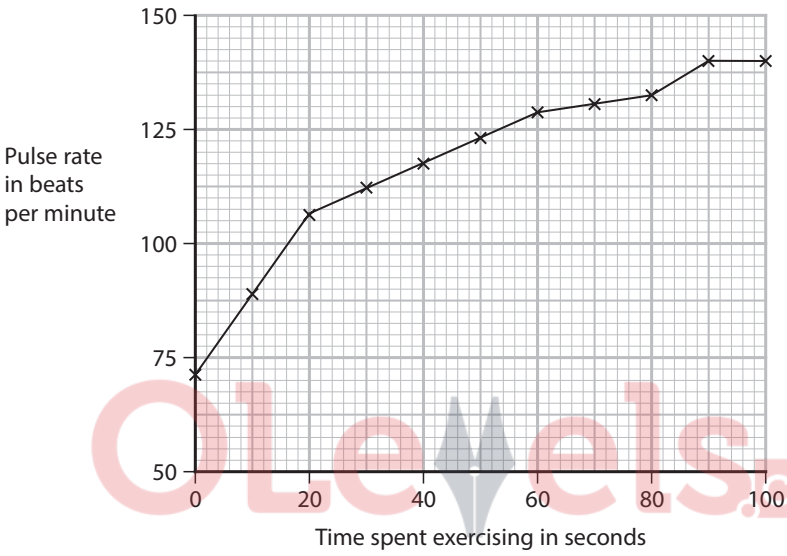
1. Question 5 · part a · 5 marks

June 2023 Paper 2

Worked-example retry · Attempt part a only.

(b) A student investigates the effect of time spent exercising on pulse rate.

The graph shows the student's results.



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1. Question 5 · part a · 5 marks · Working and answer

Working and answer

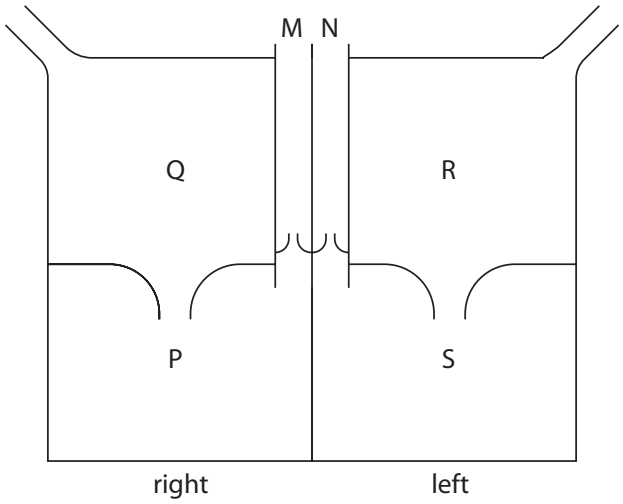
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2. Question 1 · part a.iii · 2 marks

January 2022 Paper 1

Further practice · Attempt part a.iii only.

1 The diagram shows a human heart in simple form.



(a) (i) Which of these is the name of blood vessel M?

(1)

- ☒ **A** aorta
- ☒ **B** pulmonary artery
- ☒ **C** pulmonary vein
- ☒ **D** vena cava

(ii) Which of these is the name of blood vessel N?

(1)

- ☒ **A** aorta
- ☒ **B** pulmonary artery
- ☒ **C** pulmonary vein
- ☒ **D** vena cava

(iii) Draw arrows on the diagram to show the flow of blood through the heart.

(2)

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2. Question 1 · part a.iii · 2 marks

January 2022 Paper 1

Further practice · Attempt part a.iii only.

(iv) Explain why the blood can only flow in the direction of the arrows.

(2)

(b) The width of the walls of each of the four chambers is measured.

These are the results.

0.2 cm 1.0 cm 1.5 cm 0.2 cm

(i) Complete the table by putting the correct width next to the letter of each chamber.

(3)

Chamber	Width in cm
P	
Q	
R	
S	

(ii) Explain the differences in the widths.

(3)

(Total for Question 1 = 12 marks)

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2. Question 1 · part a.iii · 2 marks · Working and answer

Working and answer

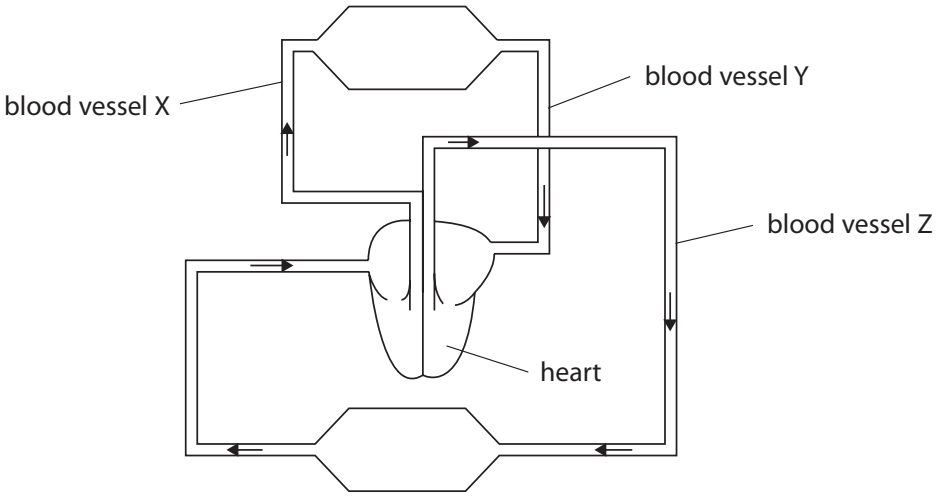
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3. Question 1 · part b · 5 marks

June 2022 Paper 2

Further practice · Attempt part b only.

1 The diagram shows a simple version of the human circulatory system.



(a) Name blood vessel X. **(1)**

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3. Question 1 · part b · 5 marks

June 2022 Paper 2

Further practice · Attempt part b only.

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(b) The box gives words about the circulatory system.

body	oxygenated	lungs	ventricle
	deoxygenated	atrium	

Complete the sentences using words from the box.

(5)

Blood vessel Y transports ..... blood from the

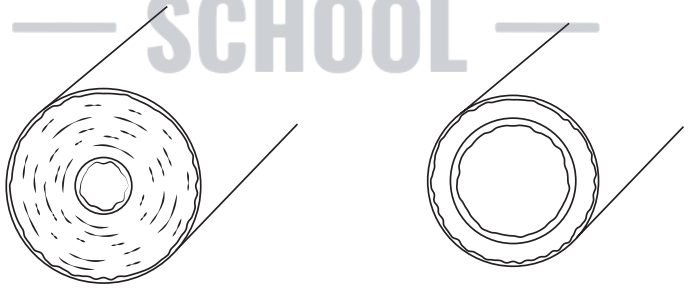
..... to the left ..... of the heart.

Blood vessel Z transports blood from the left ..... of the heart

to the .....

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(c) The diagram shows the structure of blood vessel X and blood vessel Y.



blood vessel X blood vessel Y

Compare the structure of blood vessel X with the structure of blood vessel Y.

(3)

.....

.....

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.....

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(Total for Question 1 = 9 marks)

3. Question 1 · part b · 5 marks · Working and answer

Working and answer

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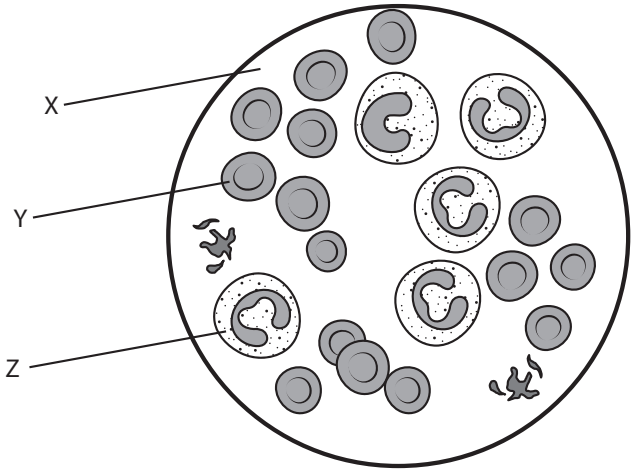
4. Question 2 · part c.ii · 4 marks

June 2024 Paper 1R

Further practice · Attempt part c.ii only.

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2 (a) The diagram shows components, X, Y and Z, in a sample of human blood.



(i) Name components X, Y and Z.

(3)

X .....

Y .....

Z .....

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(ii) Describe the function of component Y.

(2)

.....

.....

.....

.....

(iii) A sample of blood from a patient is analysed in a hospital.

A scientist finds that the number of component Z in the blood sample is higher than normal.

State a possible reason for the increase in the number of component Z.

(1)

.....

.....

4. Question 2 · part c.ii · 4 marks

June 2024 Paper 1R

Further practice · Attempt part c.ii only.

(b) Blood is transported around the body inside blood vessels.

Draw **one** straight line from each blood vessel to a feature of the blood vessel.

(2)

Blood vessel	Feature of blood vessel
artery	contains valves to prevent backflow of blood
vein	has muscular walls to maintain high blood pressure
capillary	has thin walls for exchange of materials

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4. Question 2 · part c.ii · 4 marks

June 2024 Paper 1R

Further practice · Attempt part c.ii only.

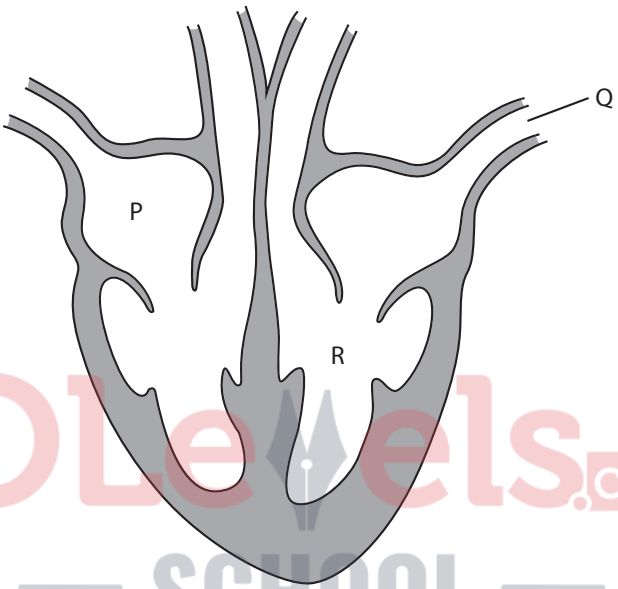
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(c) The heart pumps blood around the body.

The diagram shows a human heart.



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(i) The heart is divided into two halves.

Each half is divided into two chambers.

Name chamber R.

(1)

(ii) Describe the route that the blood travels from part P to part Q.

(4)

(Total for Question 2 = 13 marks)

4. Question 2 · part c.ii · 4 marks · Working and answer

Working and answer

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5. Question 5 · part b · 3 marks

June 2024 Paper 2R

Further practice · Attempt part b only.

5 Read the passage below. Use the information in the passage and your own knowledge to answer the questions that follow.

The heart is a double-pump that is part of the human circulatory system. It helps ensure that the blood and its components reach all cells in the body.

Some components of the blood are made in the bone marrow. An average adult with a mass of 70 kg has 5.5 dm^3 of blood. About 45% of the blood is made up of different cells. There are several different types of cell in the blood but red blood cells are the most common. In males, 1.0 dm^3 of blood contains 5.9×10^{12} red blood cells. In females the same volume of blood contains 5.5×10^{12} red blood cells. Haemoglobin makes up a large part of red blood cells. This protein is vital to the function of red blood cells.

Red blood cells are transported around the body in a liquid known as plasma. Only 10% of blood plasma is made up of components that are used or produced by body cells. The rest of the plasma is made up of water. Some components of blood plasma are used by body cells in chemical reactions, some act as messengers or have a role in blood clotting and others are excreted from the body as waste.

(a) Name the main tissue that makes up the walls of the heart. **(1)**

(b) Explain why the heart is known as a double-pump. (Lines 1 to 2) **(3)**

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5. Question 5 · part b · 3 marks

June 2024 Paper 2R

Further practice · Attempt part b only.

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(c) Red blood cells are the most common type of cell found in the blood.

Which of these are also types of cell found in the blood?

(1)

- ☒ **A** phagocytes and platelets
- ☒ **B** antibodies and hormones
- ☒ **C** lymphocytes and phagocytes
- ☒ **D** platelets and lymphocytes

(d) (i) Name a component of the blood plasma that acts as a messenger. (Lines 12 to 14)

(1)

.....

(ii) Name a component of the blood that is involved in blood clotting.

(1)

.....

(e) Calculate the number of red blood cells found in the blood of an adult male with a mass of 95 kg. (Lines 3 to 7)

Give your answer in standard form.

(3)

number of red blood cells = ...

(Total for Question 5 = 10 marks)

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5. Question 5 · part b · 3 marks · Working and answer

Working and answer

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6. Question 1 · part a · 8 marks

June 2025 Paper 1R

Further practice · Attempt part a only.

1 (a) The sentences are about the heart.

Complete the sentences by using words or numbers from the box.

(8)

80	100	120	atria	body	diastolic	elastic tissue
lungs	muscle	systolic	tendon	vena cava	ventricles	

The heart consists of four chambers.

The two upper chambers are called ..... .

The left upper chamber receives blood from the ..... whilst the right upper chamber receives blood from the body through the ..... .

The two lower chambers are called ..... .

The left lower chamber pumps blood to the ..... .

This left lower chamber has a very thick wall made of ..... .

When the chamber contracts, it generates a pressure known as the ..... pressure.

The average value for this pressure is about ..... mmHg.

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P 7 4 6 2 8 A 0 2 2 0

6. Question 1 · part a · 8 marks

June 2025 Paper 1R

Further practice · Attempt part a only.

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(b) Cardiac output is the volume of blood pushed out of the heart in one minute.

Cardiac output is calculated using this formula.

cardiac output = heart rate × stroke volume

The stroke volume is the volume, in dm^3, of blood pushed out during one heartbeat. The heart rate is the number of beats per minute.

Calculate the stroke volume for a person with a heart rate of 68 beats per minute and a cardiac output of 6.1 dm^3.
Give your answer in cm^3.

(3)

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stroke volume = ... cm^3

(Total for Question 1 = 11 marks)

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6. Question 1 · part a · 8 marks · Working and answer

Working and answer

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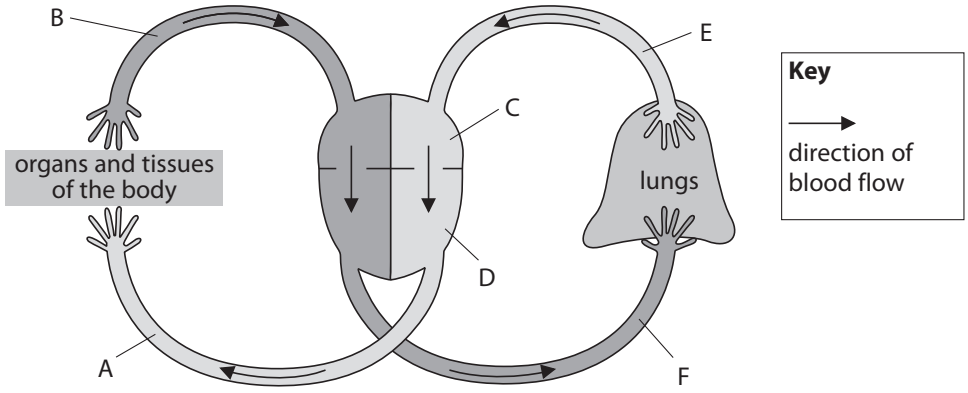
7. Question 4 · part b · 4 marks

January 2020 Paper 1

Further practice · Attempt part b only.

4 William Harvey was the first person to describe the circulation of blood around the body.

The diagram shows a model of the human circulatory system based on the research carried out by Harvey.



(a) Harvey suggested that each side of the heart was divided into two chambers.

State the names of chambers C and D.

(2)

chamber C .....

chamber D .....

(b) Harvey also suggested that blood is transported to and from the heart in different types of vessel.

Explain how the diagram supports Harvey's suggestion.

(4)

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P 5 9 9 0 9 A 0 1 0 2 8

7. Question 4 · part b · 4 marks

January 2020 Paper 1

Further practice · Attempt part b only.

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(c) Before Harvey's discoveries some scientists thought that the blood was circulated around the body by the liver.

State why this idea is not correct.

(1)

(d) Harvey's research did not include capillaries.

Describe the function of capillaries.

(2)

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(Total for Question 4 = 9 marks)

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7. Question 4 · part b · 4 marks · Working and answer

Working and answer

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8. Question 3 · part a.iii · 3 marks

November 2023 Paper 1

Further practice · Attempt part a.iii only.

3 The table shows the flow of blood to different organs in a person at rest and during exercise.

	Blood flow in cm^3 per minute	
	at rest	during exercise
Heart	250	750
Skeletal muscles	1000	12 500
Skin	400	1900
Gut	1400	600
Brain	750	750
Total flow	5600	17 500

(a) (i) Give the ratio of the blood flow to the heart when at rest and during exercise.

Give your answer in the form 1 : n

(2)

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Ratio = ... : .....

(ii) Calculate the percentage increase in the total flow of blood between rest and exercise.

Give your answer to two significant figures.

(3)

percentage increase = ... %

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8. Question 3 · part a.iii · 3 marks

November 2023 Paper 1

Further practice · Attempt part a.iii only.

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(iii) Explain why the total blood flow at rest is greater than the flow to the heart, skeletal muscle, skin, gut and brain added together.

(3)

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(b) (i) Describe the main changes between the blood flow values at rest and during exercise.

(3)

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(ii) Explain the change in blood flow rate to the skin when at rest and during exercise.

(4)

(Total for Question 3 = 15 marks)

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8. Question 3 · part a.iii · 3 marks · Working and answer

Working and answer

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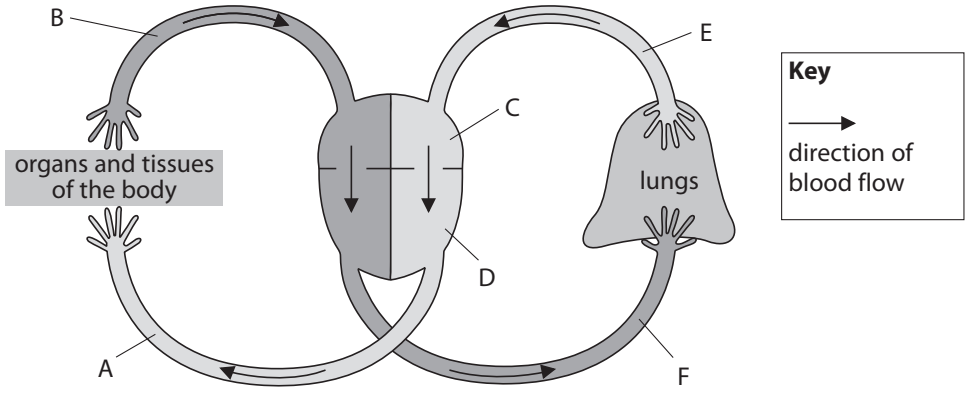
9. Question 4 · part c · 1 mark

January 2020 Paper 1

Further practice · Attempt part c only.

4 William Harvey was the first person to describe the circulation of blood around the body.

The diagram shows a model of the human circulatory system based on the research carried out by Harvey.



(a) Harvey suggested that each side of the heart was divided into two chambers.

State the names of chambers C and D.

(2)

chamber C .....

chamber D .....

(b) Harvey also suggested that blood is transported to and from the heart in different types of vessel.

Explain how the diagram supports Harvey's suggestion.

(4)

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9. Question 4 · part c · 1 mark

January 2020 Paper 1

Further practice · Attempt part c only.

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(c) Before Harvey's discoveries some scientists thought that the blood was circulated around the body by the liver.

State why this idea is not correct.

(1)

.....

.....

(d) Harvey's research did not include capillaries.

Describe the function of capillaries.

(2)

.....

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(Total for Question 4 = 9 marks)

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Working and answer